

Design of a Solar Array Based Single Input Multi Output Converter

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Abstract

This report presents the design and simulation of a solar array based single input multi output converter. The proposed system uses a photovoltaic array as the input source and generates multiple isolated DC bus voltages using a multi-winding transformer based converter structure. Averaged converter models are used for the simulation in order to reduce computational time and improve simulation stability. Although high-fidelity switching models provide more accurate device-level behavior, they require significantly higher simulation time, especially when multiple converter stages and transformer windings are involved. The system is designed using a photovoltaic source, averaged power converter blocks, a multi-winding transformer, rectification stages, and an inverter with an LCL output filter.

1 Introduction

Renewable energy based power conversion systems are increasingly used in electric vehicles, microgrids, aircraft power systems, battery charging systems, and distributed energy networks. Among renewable sources, photovoltaic systems are widely used because of their modularity, low maintenance, and clean energy generation.

In many practical applications, a single input source is required to supply multiple voltage levels. For example, a system may require a high-voltage DC bus for an inverter, a medium-voltage DC bus for auxiliary loads, and a low-voltage DC bus for control electronics. Using separate converters for each output increases system size, cost, and complexity. Therefore, a single input multi output converter is a useful solution.

The objective of this project is to design a solar array based single input multi output converter using averaged converter models. The photovoltaic array acts as the main input source, while the multi-winding transformer provides isolated multiple output ports.

2 System Description

The proposed converter consists of the following major stages:

1. Photovoltaic array source
2. DC-DC converter stage
3. Multi-winding transformer
4. Multiple rectifier and DC bus stages
5. Inverter stage
6. LCL output filter
7. Control circuit

The generalized structure of the system is shown below:

PV Array $\rightarrow$ Averaged DC-DC Converter $\rightarrow$ Multi-Winding Transformer
 $\rightarrow$ Rectifier Stages $\rightarrow$ Multiple DC Buses $\rightarrow$ Inverter $\rightarrow$ LCL Filter

The use of averaged converter models allows the converter behavior to be studied without simulating every switching transition of the semiconductor devices. This is especially useful for multiport converter systems where detailed switching simulation can become computationally expensive.

3 Photovoltaic Array Design

The photovoltaic array is designed using the A2Peak Power POWER ON P220-6x10 module. The selected module has a maximum power of approximately 220 W. The array is formed using multiple parallel strings and series-connected modules per string.

Table 1: PV Array Parameters

Parameter	Value
PV module	A2Peak Power POWER ON P220-6x10
Maximum power per module	219.978 W
Cells per module	60
Open-circuit voltage, V_{oc}	36.72 V
Short-circuit current, I_{sc}	7.98 A
Voltage at maximum power point, V_{mp}	30.3 V
Current at maximum power point, I_{mp}	7.26 A
Temperature coefficient of V_{oc}	-0.35 %/°C
Temperature coefficient of I_{sc}	0.05 %/°C
Parallel strings	40
Series-connected modules per string	8
Irradiance	1000 W/m ²
Cell temperatures	45°C and 25°C

The approximate maximum power of the complete PV array is:

$$P_{array} = P_{module} \times N_s \times N_p$$

where,

$$P_{module} = 219.978W$$

$$N_s = 8$$

$$N_p = 40$$

Therefore,

$$P_{array} = 219.978 \times 8 \times 40$$

$$P_{array} \approx 70.39 \text{ kW}$$

The approximate voltage at maximum power point of one string is:

$$V_{mp,array} = V_{mp,module} \times N_s$$

$$V_{mp,array} = 30.3 \times 8 = 242.4V$$

The approximate current at maximum power point is:

$$I_{mp,array} = I_{mp,module} \times N_p$$

$$I_{mp,array} = 7.26 \times 40 = 290.4A$$

Thus, the PV array is capable of supplying a high-power DC input suitable for a multi-output converter system.

4 Multi-Winding Transformer Design

A multi-winding transformer is used to obtain multiple isolated output voltage levels from a single input source. In this project, a 1:3 multi-winding transformer configuration is used. This allows a single input winding to magnetically couple energy to three output windings.

The transformer is modeled using per-unit parameters. The multi-winding transformer allows the number of windings on the left and right side of the block to be configured according to the required number of ports.

Table 2: Multi-Winding Transformer Parameters

Parameter	Value
Transformer type	Multi-winding transformer
Configuration	1:3
Nominal power	75 kVA
Nominal frequency	50 Hz
Primary winding nominal voltage	415 V rms
Secondary winding nominal voltages	60 V, 60 V, 60 V rms
Winding resistances	0.005 pu for each winding
Winding leakage inductances	0.02 pu for each winding
Magnetization resistance, R_m	50 pu
Magnetization inductance, L_m	50 pu

The transformer provides galvanic isolation between the input source and the output buses. It also allows different voltage levels to be obtained by modifying the turns ratio of each secondary winding.

5 Averaged Converter Modeling

In this project, averaged converter models are used instead of highly detailed switching models. In a switching model, semiconductor devices such as MOSFETs, IGBTs, and diodes are switched at high frequency. This requires very small simulation step sizes, especially when high-frequency transformers and multiple output ports are present.

In an averaged converter model, the switching action is represented by averaged voltage and current relationships. This significantly reduces simulation time while still preserving the essential dynamic behavior of the converter.

The averaged model is useful for:

- Studying steady-state output voltages
- Observing converter dynamics
- Testing control loops
- Reducing simulation time
- Avoiding numerical stiffness due to high-frequency switching

However, averaged models do not show actual switching ripple, semiconductor switching losses, diode reverse recovery, or detailed electromagnetic transient behavior.

6 Output Inverter and LCL Filter

The inverter converts the regulated DC bus voltage into AC output voltage. An LCL filter is connected at the inverter output to reduce switching harmonics and improve the quality of the AC waveform.

The LCL filter parameters used in this design are shown in Table 3.

Table 3: LCL Filter Parameters

Parameter	Value
Filter inductance, L	1 mH
Filter capacitance, C	10 mF
Filter type	LCL output filter

The LCL filter attenuates high-frequency switching components and allows the inverter to supply a smoother AC voltage waveform. Since averaged converter models are used, the ripple behavior is not as detailed as in a switching model, but the filter still represents the output dynamics of the inverter stage.

7 Control Circuit

A PI controller is used in the control circuit. The controller parameters selected for the simulation are:

Table 4: PI Controller Parameters

Controller Parameter	Value
Proportional gain, K_p	0.1
Integral gain, K_i	0.001

The general form of the PI controller is:

$$G_c(s) = K_p + \frac{K_i}{s}$$

Substituting the selected values:

$$G_c(s) = 0.1 + \frac{0.001}{s}$$

The PI controller is used to regulate the converter output by reducing the error between the reference value and the measured output value.

Since the load is not connected in the present simulation, the inner current loop has not been tuned according to load requirements. The tuning of the inner current loop depends strongly on the load power, load dynamics, desired transient response, and allowable current ripple. Therefore, the present control design focuses on basic voltage regulation and system-level verification.

8 Reason for Not Using High-Fidelity Switching Models

Highly detailed switching models provide more accurate results because they include switching transitions, device-level behavior, diode recovery, parasitic effects, and ripple. However, they require a small solver step size and high computational effort.

For this project, high-fidelity switching simulation would take a large amount of time because the system includes:

- A large PV array
- Multi-output converter structure
- Multi-winding transformer
- Multiple DC bus stages
- Inverter stage
- LCL filter
- Control loops

Therefore, averaged models are selected to reduce simulation complexity and obtain faster system-level results. This approach is suitable during the initial design stage.

9 Advantages of the Converter Strategy

The proposed solar array based single input multi output converter has several advantages:

1. **Multiple voltage outputs:** A single PV source can be used to generate multiple DC bus voltages.
2. **Reduced converter count:** Instead of using separate converters for each output, one integrated converter structure can be used.
3. **Galvanic isolation:** The multi-winding transformer provides isolation between input and output stages.
4. **Compact system structure:** Multiple outputs can be obtained from a common magnetic structure.
5. **Suitable for renewable systems:** The topology is useful for PV-based standalone and grid-connected applications.

6. **Flexible voltage design:** Output voltages can be modified by changing transformer turns ratios.
7. **Improved simulation speed:** Averaged converter models reduce computational burden during system-level design.

10 Disadvantages of the Converter Strategy

The converter strategy also has some limitations:

1. **Cross-regulation issue:** In multi-output converters, variation in one output load can affect the other output voltages.
2. **Transformer design complexity:** Multi-winding transformers require careful design to minimize leakage inductance and losses.
3. **Control complexity:** Regulating multiple outputs from a single input source is more difficult than regulating a single-output converter.
4. **Unequal load sharing:** Different output ports may experience unequal loading, which can affect voltage regulation.
5. **Limited accuracy of averaged models:** Averaged models do not show switching ripple, switching losses, and detailed transient behavior.
6. **Need for proper load-based tuning:** The inner current loop and voltage loop must be tuned according to actual load requirements.
7. **Magnetic coupling effects:** Leakage inductance and parasitic capacitance can affect converter performance in practical hardware.

11 Applications

The proposed converter strategy can be used in several applications, including:

- Solar photovoltaic power conversion systems
- DC microgrids
- Electric vehicle auxiliary power supplies
- Battery charging systems
- Renewable energy based standalone power systems
- Aircraft and aerospace DC power distribution systems

- Telecom power supplies
- Industrial auxiliary power systems
- Multi-output isolated power supplies
- Hybrid energy storage systems

12 Simulation Diagram and Block Description

The simulation model of the proposed solar array based single input multi output converter is shown in Fig. 1. The complete system is built using averaged converter blocks in MATLAB/Simulink. The main blocks used in the simulation are the PV array, boost converter, single-phase inverter, multi-winding transformer, single-phase uncontrolled rectifier, and buck converter.

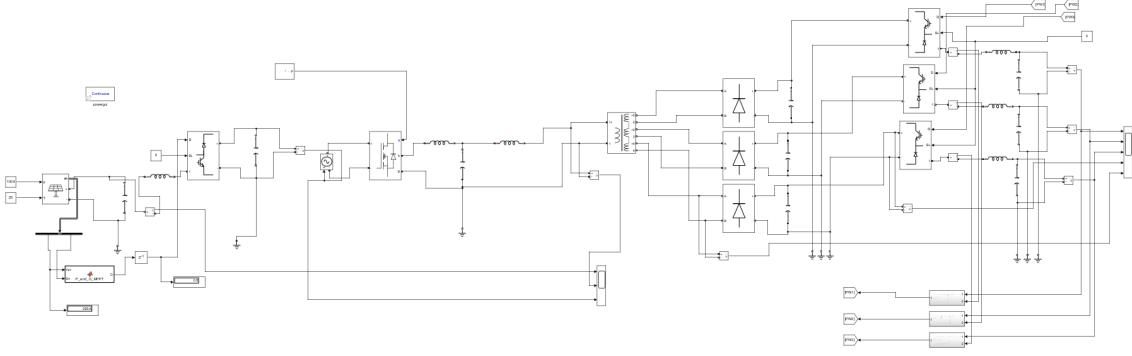


Figure 1: Simulation diagram of the solar array based single input multi output converter

The overall power flow of the system is:

PV Array $\rightarrow$ Boost Converter $\rightarrow$ Single-Phase Inverter $\rightarrow$ Multi-Winding Transformer $\rightarrow$ Rectifier Sta

12.1 PV Array

The PV array acts as the main input source of the system. It converts solar irradiance into electrical DC power. The PV array is configured using 40 parallel strings and 8 series-connected modules per string. The output voltage and current of the PV array depend on irradiance, temperature, and the operating point imposed by the power converter.

In this simulation, the PV array is connected to an MPPT-controlled boost converter. The MPPT controller adjusts the duty cycle of the boost converter so that the PV array operates close to its maximum power point.

12.2 Boost Converter

The boost converter is used to step up the PV array voltage to a higher DC-link voltage. Since the PV array voltage varies with irradiance, temperature, and loading conditions, the boost converter provides an intermediate regulated or controlled DC voltage for the next conversion stage.

For an ideal boost converter, the output voltage is given by:

$$V_{boost} = \frac{V_{pv}}{1 - D}$$

where D is the duty cycle. By changing the duty cycle, the MPPT controller controls the PV operating voltage and extracts maximum power from the solar array.

12.3 Single-Phase Inverter

The boosted DC voltage is converted into AC using a single-phase inverter. The inverter produces an AC voltage that is applied to the primary side of the multi-winding transformer.

The role of the inverter is important because the transformer requires an alternating voltage for magnetic coupling. Therefore, the inverter acts as the interface between the DC-link and the transformer stage.

12.4 Multi-Winding Transformer

The multi-winding transformer is used to obtain multiple isolated output voltages from a single input. In this project, a 1:3 multi-winding transformer is used. This means that one primary winding is used on the input side and three secondary windings are used on the output side.

The transformer provides:

- Galvanic isolation between input and output stages
- Multiple output ports from a single input source
- Voltage scaling based on turns ratio
- Common magnetic coupling for the output buses

The secondary voltages of the transformer are then supplied to rectifier stages to generate DC outputs.

12.5 Single-Phase Uncontrolled Rectifier

A single-phase uncontrolled rectifier is connected at the transformer secondary side. The rectifier converts the AC voltage from the transformer secondary winding into DC voltage.

Since an uncontrolled rectifier is used, the output voltage depends on the transformer secondary voltage, diode voltage drops, and the output filter. The rectifier output contains ripple, so a capacitor filter is used to smooth the DC voltage.

The rectifier stage performs the following functions:

- Converts transformer secondary AC voltage into DC voltage
- Provides a DC input to the downstream buck converter
- Allows each transformer secondary to form an independent DC bus path

12.6 Buck Converter

The buck converter is used after the rectifier stage to step down and regulate the DC voltage for each output bus. Each bus can be controlled to obtain a required voltage level depending on the application.

For an ideal buck converter, the output voltage is given by:

$$V_{bus} = DV_{in}$$

where V_{in} is the rectifier output voltage and D is the duty cycle of the buck converter.

The buck converter is useful because the rectifier output voltage may not directly match the required bus voltage. Therefore, the buck stage provides voltage regulation for Bus 1, Bus 2, and Bus 3.

12.7 Output DC Buses

The outputs of the buck converters form the final DC buses of the system. In the present simulation, three DC bus voltages are observed:

- Bus 1 voltage
- Bus 2 voltage
- Bus 3 voltage

Since the loads are not connected in the current simulation, the bus capacitors charge to their respective steady-state voltages. Under loaded conditions, the bus voltages would depend on the load current, buck converter duty cycle, filter parameters, and controller tuning.

12.8 Summary of Blocks Used

Table 5: Role of Major Simulation Blocks

Block	Role in the Simulation
PV Array	Acts as the renewable DC input source and supplies power to the converter system.
Boost Converter	Steps up the PV voltage and works with the MPPT controller to operate the PV array near maximum power point.
Single-Phase Inverter	Converts the boosted DC voltage into AC voltage to excite the transformer primary winding.
Multi-Winding Transformer	Provides galvanic isolation and generates multiple secondary voltages from a single primary input.
Single-Phase Uncontrolled Rectifier	Converts each transformer secondary AC voltage into DC voltage.
Buck Converter	Steps down and regulates the rectified DC voltage to obtain the required output bus voltages.
Output DC Buses	Represent the final regulated DC outputs of the single input multi output converter.

13 Results and Discussion

The simulation of the solar array based single input multi output converter was carried out using averaged converter models. The use of averaged models allowed faster simulation while still capturing the major dynamic behavior of the PV source, boost stage, transformer, rectifier, DC buses, inverter, and output filter.

The MPPT controller was used to extract maximum power from the PV array. The PV voltage and PV current were monitored to verify the operating behavior of the solar array. The boost converter output voltage was observed to confirm voltage step-up operation before feeding the multi-output conversion stage.

The important simulation outputs observed are listed below:

1. PV characteristics
2. PV voltage
3. PV current
4. Boost converter output voltage
5. AC voltage
6. Transformer secondary voltage
7. Rectifier output voltage

8. Bus 1 voltage
9. Bus 2 voltage
10. Bus 3 voltage

13.1 PV Characteristics

The PV characteristics were plotted to observe the current-voltage and power-voltage behavior of the solar array under the selected irradiance and temperature conditions. The PV array was configured with 40 parallel strings and 8 series-connected modules per string. The characteristics verify the maximum power operating region of the solar array.

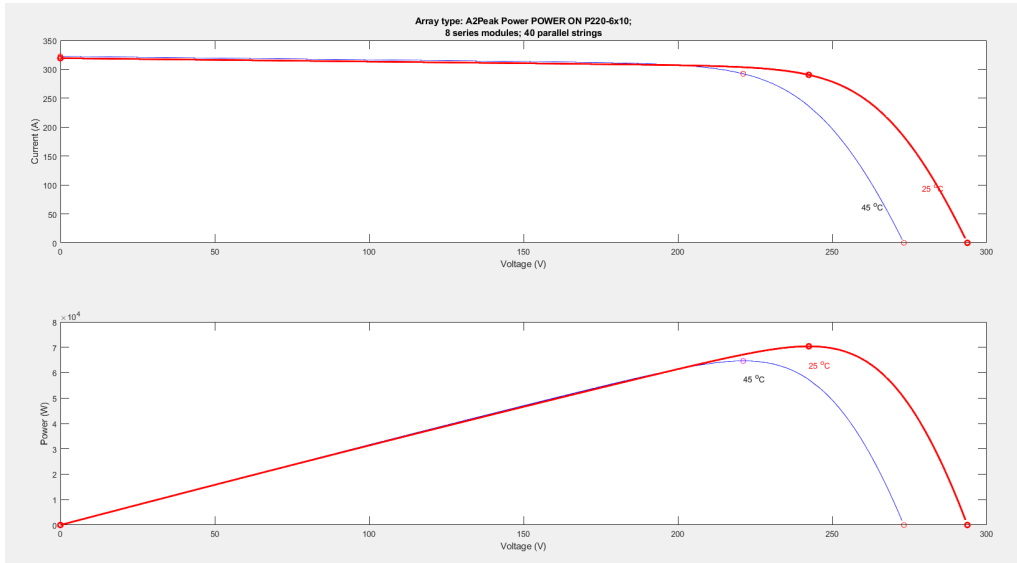


Figure 2: PV array I-V and P-V characteristics

The PV curve shows the nonlinear behavior of the solar array. The maximum power point is located near the knee region of the P-V curve. The MPPT controller operates by adjusting the converter duty cycle so that the PV array operates close to this maximum power point.

13.2 PV Voltage

The PV voltage waveform is used to verify whether the PV array is operating near the desired maximum power point voltage. For the selected PV configuration, the approximate voltage at maximum power point is:

$$V_{mp,array} = 30.3 \times 8 = 242.4 \text{ V}$$



Figure 3: PV voltage waveform

The PV voltage settles according to the duty cycle generated by the MPPT controller. Small variations may be observed due to perturbation in the MPPT algorithm.

13.3 PV Current

The PV current waveform represents the current supplied by the solar array to the converter. For the selected configuration, the approximate current at maximum power point is:

$$I_{mp,array} = 7.26 \times 40 = 290.4 \text{ A}$$

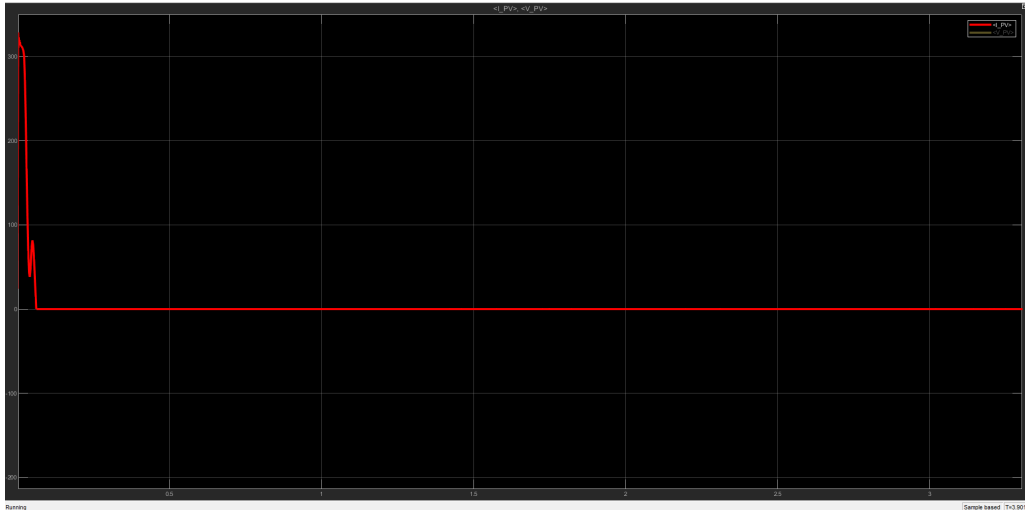


Figure 4: PV current waveform

The PV current depends on irradiance, temperature, operating voltage, and converter duty cycle. Since MPPT is used, the PV current changes as the controller searches for the maximum power operating point.

13.4 Boost Converter Output Voltage

The boost converter is used to increase the PV array voltage to a higher DC-link level. The output voltage of the boost converter is related to the input voltage and duty cycle by the ideal boost converter equation:

$$V_{boost} = \frac{V_{pv}}{1 - D}$$

where D is the duty cycle.

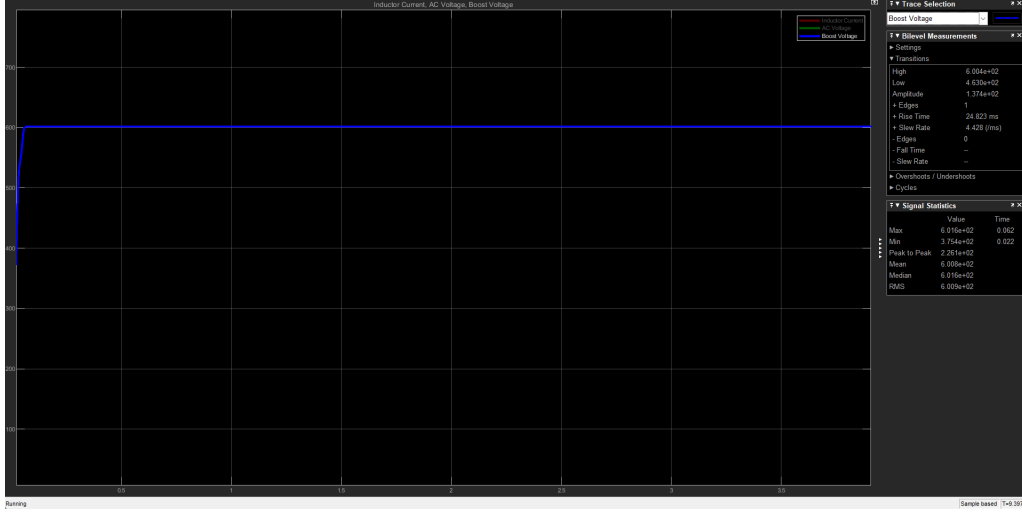


Figure 5: Boost converter output voltage

The boost voltage increases from its initial value and settles to the required DC-link level. The settling behavior depends on the converter parameters, control gains, and output capacitance.

13.5 AC Voltage

The boosted DC voltage is converted into AC using an inverter. The AC voltage waveform is observed before or after the LCL filter depending on the measurement point in the simulation.

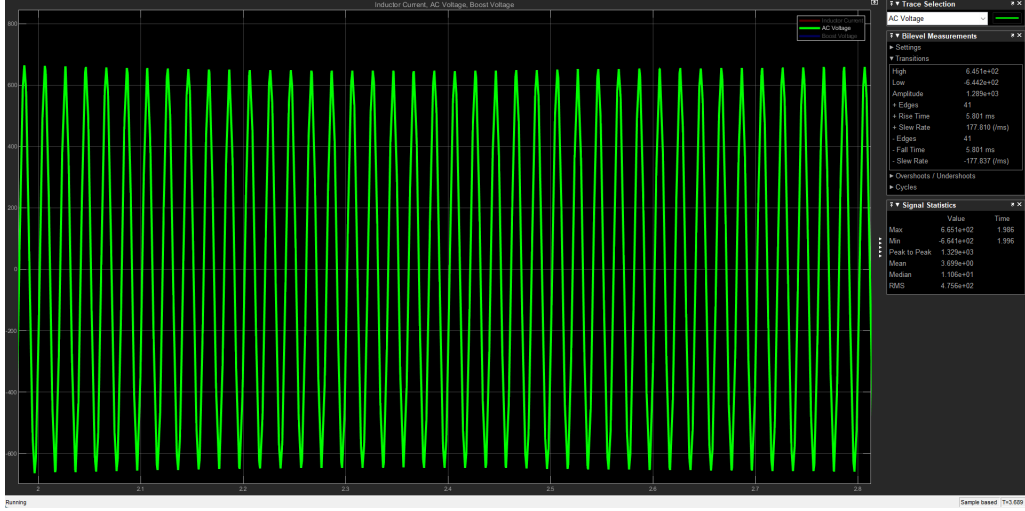


Figure 6: AC voltage waveform

The LCL filter reduces the high-frequency components of the inverter output voltage. Since averaged converter models are used, the detailed switching ripple is not fully represented, but the fundamental AC voltage behavior can still be studied.

13.6 Transformer Secondary Voltage

The multi-winding transformer provides isolated secondary voltages for the different output ports. A 1:3 multi-winding transformer configuration is used in this design.

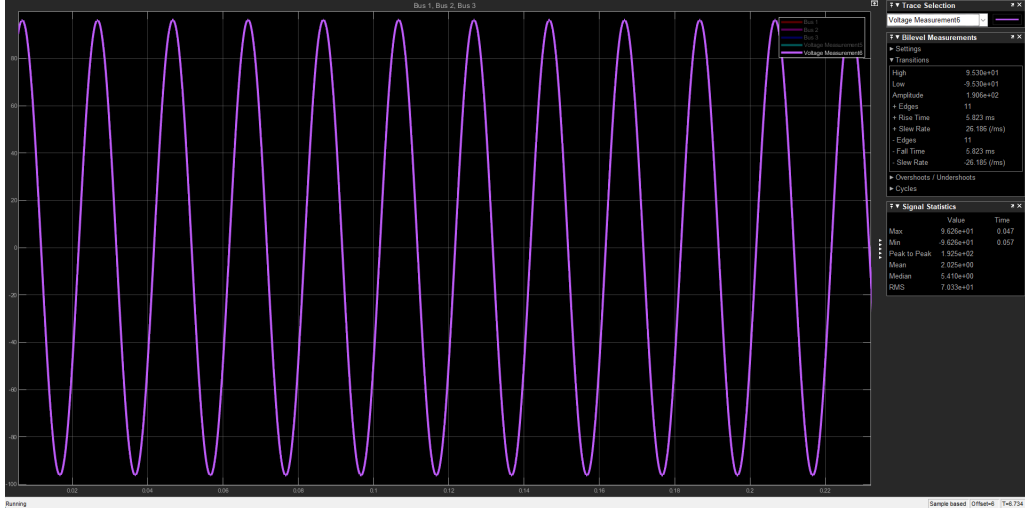


Figure 7: Transformer secondary voltage

The secondary voltage depends on the transformer turns ratio and the applied primary-side AC voltage. Since multiple secondary windings are used, each output port can be designed for a required voltage level.

13.7 Rectifier Output Voltage

The transformer secondary AC voltages are rectified to obtain DC voltages. The rectifier output voltage is observed before the final DC bus filtering stage.



Figure 8: Rectifier output voltage

The rectifier output contains a DC component along with ripple. The output capacitor reduces this ripple and provides a smoother DC voltage to the bus.

13.8 Bus 1 Voltage

Bus 1 represents the first DC output obtained from the multi-output converter.

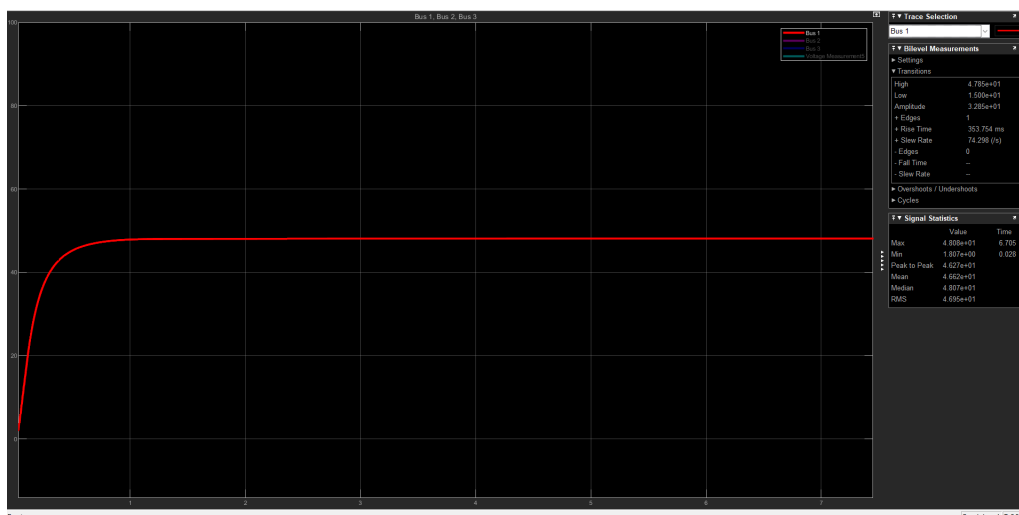


Figure 9: Bus 1 voltage

The Bus 1 voltage settles after the initial transient. Since no load is connected, the bus capacitor charges to its steady value and the current demand becomes very small.

13.9 Bus 2 Voltage

Bus 2 represents the second isolated DC output.

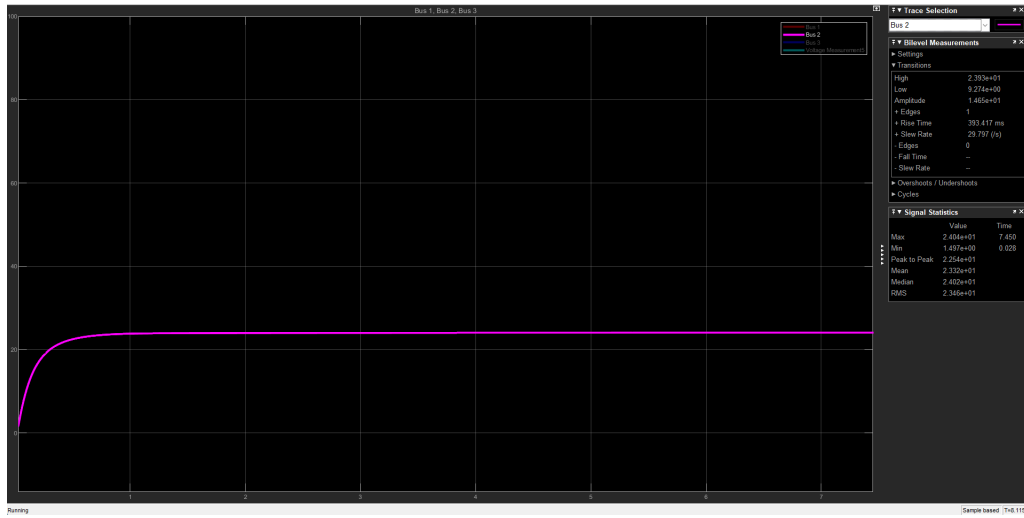


Figure 10: Bus 2 voltage

The Bus 2 voltage depends on the corresponding transformer secondary winding voltage, rectifier stage, and filter capacitor. In a loaded condition, this bus voltage would vary depending on the load connected to that port.

13.10 Bus 3 Voltage

Bus 3 represents the third isolated DC output.

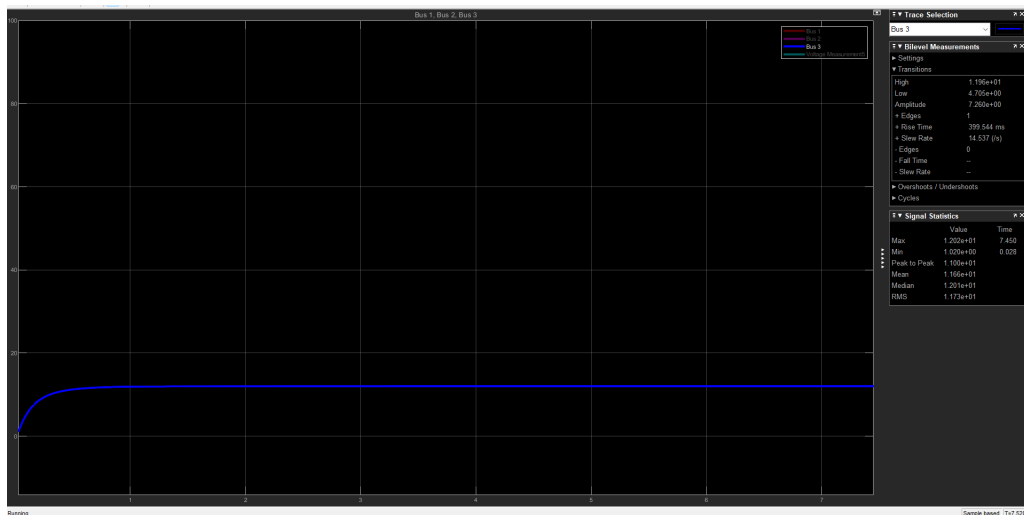


Figure 11: Bus 3 voltage

The Bus 3 voltage confirms the multi-output capability of the proposed converter. Since the present simulation does not include load connection, the output bus voltage behavior mainly represents capacitor charging and steady-state voltage formation.

13.11 Discussion

The simulation results show that the PV array, MPPT controller, boost converter, multi-winding transformer, rectifier stages, and multiple DC buses operate together as a single input multi output power conversion system. The MPPT controller adjusts the operating point of the PV array to extract maximum available power. The boost converter increases the PV voltage to a suitable DC-link level. The inverter and transformer stage provide isolated output voltages, which are then rectified to form multiple DC buses.

Since the load is not connected in the present simulation, the output bus voltages settle after the capacitors are charged. Under this condition, the inner current loop has not been tuned based on load requirements. In a practical loaded system, each bus would require load-dependent control tuning, voltage regulation, and current limiting.

14 Comparison of Expected and Actual Simulation Results

The expected and actual simulation values are compared in Table 6. The percentage error is calculated using:

$$\% \text{ Error} = \frac{\text{Actual Value} - \text{Expected Value}}{\text{Expected Value}} \times 100$$

Table 6: Comparison of Expected and Actual Values

Parameter	Expected Value	Actual Value	Percentage Error
PV voltage	242 V	293.8 V	21.40%
PV current	290.4 A	330 A initially	13.64%
PV current at steady state	290.4 A	0 A	-100%
Boost converter voltage	600 V	600 V	0%
AC voltage	415 V	475.6 V	14.60%
Transformer secondary voltage, winding 1	60 V rms	70.33 V rms	17.22%
Transformer secondary voltage, winding 2	60 V rms	70.33 V rms	17.22%
Transformer secondary voltage, winding 3	60 V rms	70.33 V rms	17.22%
Diode bridge rectifier average output voltage	99 V	106 V	7.07%
Bus 1 voltage	48 V	46.95 V	2.1875%
Bus 2 voltage	24 V	23.46	2.25%
Bus 3 voltage	12 V	11.73 V	2.25%

14.1 Reason for PV Current Reducing to Zero

The expected PV current was calculated from the rated maximum power point current of the PV array:

$$I_{mp,array} = 40 \times 7.26 = 290.4 \text{ A}$$

In the simulation, the PV current initially reaches approximately 330 A during the transient period. However, it gradually reduces and settles close to zero at steady state.

This occurs because no load is connected at the converter output. During the initial transient, the PV array supplies current to charge the DC-link capacitors and output bus capacitors. Once these capacitors become fully charged, the converter no longer has a continuous power demand. As a result, the inductor current in the boost converter dies down and the PV current also reduces to nearly zero.

Therefore, the steady-state PV current of zero does not indicate failure of the PV array or MPPT controller. It is mainly due to the no-load condition of the system. Under loaded operation, the output buses would continuously draw power, and the PV current would settle to a non-zero value depending on the load power and irradiance level.

15 Conclusion

A solar array based single input multi output converter has been designed and simulated using averaged converter models. The PV array consists of 40 parallel strings and 8 series-connected modules per string, providing an approximate maximum power of 70.39 kW. A 1:3 multi-winding transformer is used to generate multiple isolated output buses from a single input source.

The use of averaged converter models reduces simulation time and allows faster system-level analysis. An inverter with an LCL output filter is included to obtain AC output. A PI controller with $K_p = 0.1$ and $K_i = 0.001$ is used for basic control operation. Since the load is not connected, the inner current loop has not been tuned for load-specific requirements.

The proposed converter is useful for renewable energy systems requiring multiple isolated voltage levels. Future work can include load connection, current loop tuning, MPPT implementation, high-fidelity switching simulation, transformer optimization, and hardware validation.

16 Future Scope

The present work can be extended in the following ways:

1. **Load connection and validation:** Different resistive, inductive, and dynamic loads can be connected to each output bus to study load regulation.
2. **Inner current loop tuning:** The inner current control loop can be tuned based on actual load power and transient response requirements.
3. **MPPT implementation:** A maximum power point tracking algorithm such as Perturb and Observe or Incremental Conductance can be added to extract maximum power from the PV array.
4. **High-fidelity switching model:** The averaged converter can be replaced with detailed MOSFET, diode, and transformer models to study switching ripple and losses.
5. **Closed-loop multi-output control:** Independent voltage regulation loops can be designed for each output bus.
6. **Transformer optimization:** The multi-winding transformer can be optimized for leakage inductance, copper loss, core loss, and thermal performance.
7. **Battery integration:** A battery energy storage system can be connected to one of the DC buses to improve power availability during low irradiance conditions.

8. **Hardware implementation:** A scaled-down laboratory prototype can be developed to validate simulation results.
9. **Efficiency analysis:** Converter efficiency can be evaluated under different irradiance, temperature, and loading conditions.
10. **Grid-connected operation:** The inverter can be connected to the grid using synchronization and current control techniques.